# Requirements Gathering

## 1. Stakeholder Analysis

| Stakeholder | Role / Description | Key Needs & Expectations |
|---|---|---|
| **End User (Patient)** | An individual seeking to understand medical symptoms, upload medical images, or find specialists | • Simple, accurate explanations from trusted medical sources.<br>• Ability to upload images (X-rays, skin photos) and receive a text description.<br>• Receive a list of top-3 rated local specialists based on symptoms.<br>• Easy-to-use, secure interface.<br>• Assurance that personal medical data is not stored permanently. |
| **Physician (optional secondary user)** | A doctor who may use the system to educate patients or quickly access medical summaries. | • Accurate, up-to-date information from authoritative textbooks.<br>• Clear disclaimer that the system does not provide final diagnoses.<br>• Ability to recommend the tool to patients without fear of misinformation. |
| **Development Team** | 6 members (Mohamed, Salma, Rahma, Ahmed, Marihan, Ali) responsible for design, implementation, and deployment. | • Clear, well-defined requirements.<br>• Precise API documentation for frontend/backend integration.<br>• Telemetry logs to monitor performance and errors.<br>• A realistic timeline with milestones (already provided). |

# 2. Use Cases

## **Use Case 1:** Text-Based Medical Query

**Actor:** End user
**Preconditions**: User has internet access; system is running.
**Basic Flow:**

1. User types a question (e.g., "What causes abdominal pain after eating?").
2. LangGraph routes the request to Hybrid Search in Pinecone.
3. System retrieves top-3 relevant chunks from medical textbooks.
4. Llama 3.2 generates an answer using only the retrieved context.
5. UI displays the answer together with a safety disclaimer.

**Alternative Flow:** If no relevant context is found, the system responds: "I do not have enough information to answer this safely."

## **Use Case 2:** Medical Image Upload & Analysis

**Actor:** End user
**Preconditions**: Image format JPG/PNG, size < 5 MB
**Basic Flow:**

1. User uploads an image (e.g., skin rash).
2. Vision LLM analyses the image and produces a text description (e.g., "red, raised rash on forearm").
3. UI shows the description and asks user to confirm or edit.
4. After confirmation, LangGraph uses the (edited) text to search the medical database.
5. System displays the medical explanation and doctor recommendations.

# 2. Use Cases

## **Use Case 3:** Get Doctor Recommendations

**Actor:** End user (after a medical query)

**Basic Flow:**

1. System extracts a probable symptom or condition (e.g., "Eczema").
2. Calls fetch_doctor_recommendations tool, mapping symptom to specialty (e.g., "Dermatology").
3. Queries the PostgreSQL/MySQL database for real doctor profiles.
4. UI renders cards with: doctor name, specialty, rating (1-5), address, phone number.

# 3. Functional Requirements

| ID | Requirement | Priority |
|---|---|---|
| **FR-01** | **Doctor Recommendation Engine**: Map extracted symptoms to a medical specialty, then retrieve the top-3 highest-rated local specialists from the SQL database. | High |
| **FR-02** | **Embedding Generation:** Convert text chunks into vector embeddings and store them in Pinecone (or ChromaDB). | High |
| **FR-03** | **Hybrid Search:** Support both vector similarity search and BM25 keyword-based search for retrieving relevant chunks. | High |
| **FR-04** | **Vision LLM Integration:** Accept user-uploaded medical images and return a textual description of the image content. | High |
| **FR-05** | **LangGraph State Machine:** Automatically route incoming requests (casual chat, medical query, image upload) to the appropriate tool or function. | High |
| **FR-06** | **Grounded Response Generation:** The LLM must only answer using context retrieved from the Vector DB. If no relevant context exists, respond with a safe default message. | High |
| **FR-07** | **Doctor Recommendation Engine:** Map extracted symptoms to a medical specialty, then retrieve the top-3 highest-rated local specialists from the SQL database. | High |
| **FR-08** | **Streamlit UI:** Provide a chat interface, image uploader, doctor recommendation cards, and a permanent safety disclaimer. | High |
| **FR-09** | **FastAPI Backend Endpoints:** Expose REST endpoints for text search, image analysis, doctor fetching, and (optional) chat history logging. | Medium |
| **FR-10** | **Telemetry & Logging:** Log Vector DB latency, Vision LLM processing time, and tool-call errors. | Medium |
| **FR-11** | **Safety Disclaimer:** Display an unclosable banner on every page: "This system is for educational purposes only – not a substitute for professional medical advice." | High |

# 4. Non-functional Requirements

| ID | Category | Requirement | Target |
|---|---|---|---|
| **NFR-01** | **Performance – Retrieval** | Hybrid search query time (Pinecone) | < 1.5 sec |
| **NFR-02** | **Performance – End-to-end text** | Total time from user text query to start of response | < 4 sec |
| **NFR-03** | **Performance – Multimodal** | Total time for image upload → description → search → response | < 12 sec |
| **NFR-04** | **Performance – Doctor cards** | Time to fetch and render doctor cards in UI | < 1 sec |
| **NFR-05** | **Accuracy – Context relevance** | % of queries where retrieved chunks directly answer the question | ≥ 90% |
| **NFR-06** | **Accuracy – Recommendation** | % of correct symptom-to-specialty mappings | ≥ 95% |
| **NFR-07** | **Availability** | System uptime during testing and deployment phases | ≥ 99.9% |
| **NFR-08** | **Reliability – API errors** | % of failed API calls (timeout or bad data) | < 2% |
| **NFR-09** | **Usability – Task success** | % of test users who complete (upload image + get recommendation) without assistance | ≥ 85% |
| **NFR-10** | **Security – Data privacy** | Uploaded medical images must not be stored permanently; deleted after session ends. | Mandatory |
| **NFR-11** | **Security – API authentication** | Basic API key or simple token protection for deployed endpoints | Recommended |
| **NFR-12** | **Maintainability – Code quality** | Code must be commented, follow PEP8, and be modular/reusable. | Mandatory |
| **NFR-13** | **Scalability (future)** | Design should allow scaling Vector DB to 1M chunks without full rebuild. | Not required now, but considered in design |